

## ANGOULEME-COGNAC AP, France

WMO: 074200

Lat: 45.7279N

Lon: 0.2205E

Elev: 133

StdP: 99.74

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -4.9       | -2.8 | -9.8                          | 1.7 | -3.0 | -6.2 | 2.3 | -1.0 | 10.4                  | 10.1 | 9.3 | 9.3  | 3.4                   | 10   | 0.492 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.9                   | 33.0            | 20.5 | 30.8 | 19.7 | 28.9 | 19.1 | 21.6                | 29.9 | 20.8 | 28.6 | 19.9 | 27.1 | 3.2                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.0                            | 14.0 | 24.7 | 18.1 | 13.2 | 23.5 | 17.2 | 12.5 | 22.5 | 63.6          | 29.9 | 60.5 | 28.7 | 57.7 | 27.0 | 26.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 8.9               | 7.9  | 7.0 | -6.4                       | 36.7 | 3.3                | 1.5 | -8.8                                               | 37.7 | -10.7      | 38.6 | -12.5      | 39.4 | -14.9      | 40.5 |    |    |
|                   |      |     | -7.1                       | 23.5 | 3.2                | 1.3 | -9.4                                               | 24.4 | -11.2      | 25.2 | -13.0      | 25.9 | -15.3      | 26.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.8   | 5.4   | 5.8   | 9.3   | 11.7  | 14.5  | 18.4  | 19.9  | 20.3  | 17.6  | 13.9  | 9.5   | 6.9  |
|                                                                                  | DBStd       | 6.41   | 3.61  | 4.39  | 3.40  | 3.32  | 3.59  | 4.24  | 3.39  | 3.60  | 3.38  | 3.82  | 3.91  | 3.76 |
|                                                                                  | HDD10.0     | 521    | 146   | 124   | 53    | 16    | 5     | 2     | 0     | 0     | 1     | 9     | 57    | 107  |
|                                                                                  | HDD18.3     | 2267   | 401   | 351   | 279   | 200   | 129   | 49    | 21    | 20    | 53    | 143   | 265   | 356  |
|                                                                                  | CDD10.0     | 1547   | 3     | 6     | 33    | 68    | 143   | 254   | 308   | 319   | 228   | 131   | 42    | 11   |
|                                                                                  | CDD18.3     | 252    | 0     | 0     | 0     | 2     | 9     | 52    | 71    | 81    | 31    | 7     | 0     | 0    |
|                                                                                  | CDH23.3     | 2752   | 0     | 0     | 1     | 32    | 114   | 591   | 806   | 870   | 298   | 40    | 0     | 0    |
|                                                                                  | CDH26.7     | 952    | 0     | 0     | 0     | 3     | 19    | 213   | 287   | 351   | 74    | 5     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.3    | 3.4   | 3.8   | 3.9   | 3.5   | 3.4   | 3.2   | 3.1   | 2.8   | 3.0   | 3.1   | 3.4   | 3.5  |
| Precipitation                                                                    | PrecAvg     | 824    | 86    | 66    | 63    | 64    | 73    | 60    | 51    | 60    | 60    | 73    | 89    | 80   |
|                                                                                  | PrecMax     | 974    | 192   | 126   | 171   | 187   | 126   | 97    | 82    | 128   | 152   | 123   | 172   | 171  |
|                                                                                  | PrecMin     | 627    | 25    | 16    | 20    | 12    | 14    | 8     | 11    | 7     | 29    | 32    | 27    | 22   |
|                                                                                  | PrecStd     | 107    | 44    | 36    | 38    | 42    | 31    | 24    | 23    | 39    | 30    | 28    | 41    | 38   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 14.1  | 18.2  | 22.9  | 27.0  | 28.9  | 35.6  | 35.8  | 35.2  | 31.2  | 26.9  | 20.2  | 15.8 |
|                                                                                  |             | MCWB   | 11.3  | 10.7  | 12.7  | 15.8  | 18.9  | 22.1  | 21.2  | 20.7  | 17.8  | 17.1  | 15.4  | 11.9 |
|                                                                                  | 2%          | DB     | 12.8  | 15.3  | 20.1  | 23.8  | 26.2  | 31.2  | 32.1  | 33.0  | 28.9  | 24.0  | 18.1  | 14.1 |
|                                                                                  |             | MCWB   | 11.1  | 10.2  | 11.9  | 14.1  | 17.2  | 20.3  | 20.8  | 20.2  | 17.9  | 17.1  | 13.8  | 11.3 |
|                                                                                  | 5%          | DB     | 11.8  | 13.1  | 18.0  | 21.1  | 24.1  | 29.1  | 29.8  | 30.8  | 26.8  | 21.8  | 16.6  | 13.0 |
|                                                                                  |             | MCWB   | 10.4  | 9.3   | 11.5  | 13.0  | 16.6  | 19.6  | 19.4  | 20.0  | 17.6  | 16.7  | 13.5  | 11.2 |
|                                                                                  | 10%         | DB     | 10.7  | 11.2  | 15.8  | 18.8  | 22.1  | 26.9  | 27.8  | 28.1  | 24.9  | 19.9  | 15.0  | 12.0 |
|                                                                                  |             | MCWB   | 9.4   | 8.9   | 10.3  | 12.1  | 15.4  | 18.9  | 18.7  | 19.0  | 17.0  | 15.6  | 13.1  | 10.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.5  | 12.1  | 13.5  | 16.1  | 20.4  | 23.3  | 22.7  | 22.8  | 20.6  | 19.4  | 16.4  | 13.1 |
|                                                                                  |             | MCDB   | 13.1  | 16.5  | 20.0  | 24.3  | 27.1  | 31.7  | 31.9  | 31.1  | 26.7  | 24.0  | 18.9  | 14.0 |
|                                                                                  | 2%          | WB     | 11.5  | 11.0  | 12.6  | 14.8  | 18.7  | 21.4  | 21.4  | 21.4  | 19.4  | 18.1  | 14.9  | 12.3 |
|                                                                                  |             | MCDB   | 12.4  | 13.5  | 18.4  | 22.2  | 24.4  | 29.0  | 30.0  | 29.9  | 25.4  | 21.7  | 16.6  | 13.6 |
|                                                                                  | 5%          | WB     | 10.6  | 10.1  | 11.8  | 13.5  | 17.3  | 20.5  | 20.4  | 20.6  | 18.5  | 17.2  | 14.1  | 11.5 |
|                                                                                  |             | MCDB   | 11.6  | 12.1  | 16.8  | 19.8  | 22.4  | 28.0  | 28.3  | 28.6  | 24.0  | 20.4  | 15.8  | 13.0 |
|                                                                                  | 10%         | WB     | 9.4   | 9.3   | 10.9  | 12.5  | 16.0  | 19.2  | 19.5  | 19.6  | 17.6  | 16.2  | 13.1  | 10.7 |
|                                                                                  |             | MCDB   | 10.3  | 11.3  | 14.5  | 17.8  | 20.4  | 25.9  | 26.4  | 26.6  | 22.8  | 19.5  | 15.2  | 12.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.8   | 7.8   | 9.6   | 11.1  | 11.6  | 12.4  | 12.6  | 12.9  | 11.7  | 9.7   | 7.0   | 6.5  |
|                                                                                  |             | MCDBR  | 6.5   | 10.7  | 13.5  | 14.2  | 14.6  | 14.6  | 16.5  | 16.7  | 14.7  | 11.9  | 7.7   | 7.1  |
|                                                                                  |             | MCWBR  | 5.1   | 6.6   | 7.2   | 7.3   | 7.4   | 6.4   | 6.7   | 6.6   | 6.2   | 6.2   | 4.6   | 4.9  |
|                                                                                  | 5% WB       | MCDBR  | 5.8   | 9.2   | 11.7  | 12.8  | 13.5  | 13.8  | 15.2  | 14.2  | 12.0  | 9.8   | 7.0   | 5.9  |
| MCWBR                                                                            |             | 5.1    | 6.4   | 6.8   | 6.9   | 7.8   | 6.8   | 7.6   | 6.7   | 6.0   | 5.7   | 4.8   | 4.7   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.319  | 0.335 | 0.366 | 0.393 | 0.396 | 0.409 | 0.399 | 0.393 | 0.364 | 0.361 | 0.335 | 0.316 |      |
|                                                                                  | taud        | 2.469  | 2.412 | 2.330 | 2.251 | 2.295 | 2.293 | 2.319 | 2.351 | 2.412 | 2.434 | 2.453 | 2.465 |      |
|                                                                                  | Ebn at Noon | 786    | 841   | 859   | 864   | 872   | 859   | 863   | 855   | 853   | 799   | 755   | 746   |      |
|                                                                                  | Edh at Noon | 69     | 88    | 109   | 128   | 127   | 128   | 123   | 115   | 99    | 85    | 70    | 63    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.23   | 2.14  | 3.31  | 4.54  | 5.32  | 6.08  | 5.93  | 5.25  | 4.16  | 2.59  | 1.46  | 1.11  |      |
|                                                                                  | RadStd      | 0.14   | 0.26  | 0.41  | 0.54  | 0.46  | 0.52  | 0.38  | 0.41  | 0.27  | 0.22  | 0.18  | 0.16  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (20 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page